

Spring 2001 Mechanics Corrections

Swati Ravi

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1 Mechanics Problem 1

Equation 4 ERRATA

$$\begin{aligned}\dot{x}_D &= a\dot{\phi}_1 \cos \phi_1, \\ \dot{y}_D &= -a\dot{\phi}_1 \sin \phi_1, \\ \dot{x}_C &= b\dot{\phi}_1 \cos \phi_1 + \textcolor{red}{c}\dot{\phi}_2 \cos \phi_2, \\ \dot{y}_C &= \textcolor{red}{-b}\dot{\phi}_1 \sin \phi_1 - \textcolor{red}{c}\dot{\phi}_2 \sin \phi_2.\end{aligned}\tag{4}$$

Equation 9 ERRATA (typo on the Left Hand Side): Equations of motion are:

$$\begin{cases} (m_1 a^2 + \textcolor{red}{\mathcal{I}} + m_2 b^2) \ddot{\phi}_1 + m_2 bc \ddot{\phi}_2 \cos(\phi_1 - \phi_2) + m_2 bc \dot{\phi}_2^2 \sin(\phi_1 - \phi_2) = -(m_1 a + m_2 b) g \sin \phi_1, \\ m_2 c^2 \ddot{\phi}_2 + m_2 bc \ddot{\phi}_1 \cos(\phi_1 - \phi_2) - m_2 bc \dot{\phi}_1^2 \sin(\phi_1 - \phi_2) = -m_2 c g \sin \phi_2. \end{cases}\tag{9}$$

2 Mechanics Problem 2

Equation 16 matrix element has an error. The system of equations is represented as follows:

$$\begin{bmatrix} 2k - m\omega^2 & -k & -k \\ -k & 2k - m\omega^2 & -k \\ -k & -k & \textcolor{red}{2k} - \textcolor{red}{m}\omega^2 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\tag{16}$$